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| <b>EXP.NO: 5</b> | <b>Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.</b> |
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**AIM:**

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

**ALGORITHM:**

**1. Input:**

- The user fills out a web form with their **name** and **email**.
- The form will use both **GET** and **POST** methods for data submission.

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**2. Steps for Creating and Handling the HTTP GET and POST Requests:**

**1. Create an HTML Form:**

- Design an HTML form with two input fields: one for the **name** and one for the **email**.
- Create two buttons to allow the user to submit the form using **GET** or **POST** methods.
- Use the action attribute to define the **Servlet URL** and specify the form method (either GET or POST).

**2. Create the Servlet to Handle GET and POST Requests:**

- Write a Java Servlet class that will handle **both GET and POST requests**.
- The servlet will have two methods:
  - doGet() to handle the GET requests.
  - doPost() to handle the POST requests.
- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

### 3. Handle GET Requests in the Servlet:

- In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
- Output the data as a response, including the **name** and **email** in the URL (visible to the user).

### 4. Handle POST Requests in the Servlet:

- In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
- Output the data as a response, displaying the **name** and **email** without exposing them in the URL (more secure).

### 5. Return the Output:

- The Servlet should return an HTML page displaying the **name** and **email** entered by the user.
- For the **GET method**, the output should be visible in the URL.
- For the **POST method**, the output should be visible in the response body, not the URL.

### 6. Deploy and Test the Servlet:

- Compile the Servlet class and deploy it to a **Servlet container** (like **Apache Tomcat**).
- Test the form using **GET** and **POST** methods to see the difference in data submission.

### 7. Observe and Compare GET and POST Results:

- Test the **GET method**: The form data is sent in the URL.
- Test the **POST method**: The form data is sent in the HTTP request body, not visible in the URL.

**Output:**

GET vs POST Demo

localhost:8082/HelloWorld/form.html

### User Form

**Submit using GET**

Name:

Email:

---

**Submit using POST**

Name:

Email:

localhost:8082/HelloWorld/GetPos X

localhost:8082/HelloWorld/GetPostServlet?name=karthik+&email=karthik123%40gmail.com

### GET Method Used

**Name:** karthik

**Email:** karthik123@gmail.com

Data is visible in the URL

localhost:8082/HelloWorld/GetPos X

localhost:8082/HelloWorld/GetPostServlet

### POST Method Used

**Name:** karthik sah

**Email:** karthik456@gmail.com

Data is sent in HTTP body (not visible in URL)

**HTML File ( form.html ):**

```
<!DOCTYPE html>

<html>

<head>

  <title>GET vs POST Demo</title>

</head>

<body>

  <h2>User Form</h2>

  <!-- GET Form -->

  <form action="GetPostServlet" method="get">

    <h3>Submit using GET</h3>

    Name: <input type="text" name="name"><br><br>

    Email: <input type="email" name="email"><br><br>

    <input type="submit" value="Submit via GET">

  </form>

  <hr>

  <!-- POST Form -->

  <form action="GetPostServlet" method="post">

    <h3>Submit using POST</h3>

    Name: <input type="text" name="name"><br><br>

    Email: <input type="email" name="email"><br><br>

    <input type="submit" value="Submit via POST">

  </form>

</body>
```

</html>

### **Java File ( GetPostServlet.java )**

```
import java.io.*;

import jakarta.servlet.*;

import jakarta.servlet.http.*;

public class GetPostServlet extends HttpServlet {

    // Handle GET request

    protected void doGet(HttpServletRequest request, HttpServletResponse response)

        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");

        String email = request.getParameter("email");

        out.println("<html><body>");

        out.println("<h2>GET Method Used</h2>");

        out.println("<p><b>Name:</b> " + name + "</p>");

        out.println("<p><b>Email:</b> " + email + "</p>");

        out.println("<p>Data is visible in the URL</p>");

        out.println("</body></html>");

    }
```

```

// Handle POST request

protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    response.setContentType("text/html");

    PrintWriter out = response.getWriter();

    String name = request.getParameter("name");
    String email = request.getParameter("email");

    out.println("<html><body>");
    out.println("<h2>POST Method Used</h2>");
    out.println("<p><b>Name:</b> " + name + "</p>");
    out.println("<p><b>Email:</b> " + email + "</p>");
    out.println("<p>Data is sent in HTTP body (not visible in URL)</p>");
    out.println("</body></html>");

    }
}

```

**File ( web.xml ):**

```

<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
    https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
    version="6.0">

```

```
<servlet>

    <servlet-name>GetPostServlet</servlet-name>

    <servlet-class>GetPostServlet</servlet-class>

</servlet>


<servlet-mapping>

    <servlet-name>GetPostServlet</servlet-name>

    <url-pattern>/GetPostServlet</url-pattern>

</servlet-mapping>

</web-app>
```

### **Java Compiled File ( GetPostServlet.class ) :**

// Source code is decompiled from a .class file using FernFlower decompiler (from IntelliJ IDEA).

```
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;

public class GetPostServlet extends HttpServlet {

    public GetPostServlet() {

    }

    protected void doGet(HttpServletRequest var1, HttpServletResponse var2) throws
ServletException, IOException {

        var2.setContentType("text/html");

        PrintWriter var3 = var2.getWriter();
```

```

String var4 = var1.getParameter("name");

String var5 = var1.getParameter("email");

var3.println("<html><body>");

var3.println("<h2>GET Method Used</h2>");

var3.println("<p><b>Name:</b> " + var4 + "</p>");

var3.println("<p><b>Email:</b> " + var5 + "</p>");

var3.println("<p>Data is visible in the URL</p>");

var3.println("</body></html>");

}

```

protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws ServletException, IOException {

```

    var2.setContentType("text/html");

    PrintWriter var3 = var2.getWriter();

    String var4 = var1.getParameter("name");

    String var5 = var1.getParameter("email");

    var3.println("<html><body>");

    var3.println("<h2>POST Method Used</h2>");

    var3.println("<p><b>Name:</b> " + var4 + "</p>");

    var3.println("<p><b>Email:</b> " + var5 + "</p>");

    var3.println("<p>Data is sent in HTTP body (not visible in URL)</p>");

    var3.println("</body></html>");

}
}

```

## RESULT:

Thus, A Servlet was created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.